

Zaeem Mohtashim Khan

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I research the intersection of embodied robotics and human-aware AI, designing systems that reason under uncertainty to act, ask, or defer in ways humans can trust.

EDUCATION

Lahore University of Management Science | CGPA:3.90, Major CGPA: 3.97

September 2022 - June 2026

BSc (Hons.) Computer Science

Minor in Robotics

Graduated with High Distinction

Relevant Courses: Mobile Robotics, Deep Learning, Topics in Large Language Models, Junior Design Studio - Robotics, Distributed Systems, Operating Systems, Data Structures, Algorithms, Network Centric Computing, Databases, Introduction to Blockchain.

Lahore Grammar School Johar Town

September 2020 - May 2022

A-Level – 1A* 3A

Publications

Separating Clicks from Baits: Using Large Language Models to Detect Misleading YouTube Thumbnails

Wajiha Naveed, Muhammad Muneeb Pervez*, **Zaeem Mohtashim Khan***, Zafar Ayyub Qazi, Zartash Afzal Uzmi

ICWSM 2027 — *Equal contribution

- Co-authored a peer-reviewed paper accepted at ICWSM 2027 proposing a multi-modal LLM pipeline to detect misleading YouTube thumbnails.
- Led evaluation of open-weight, locally-run vision-language models (LLaVA-v1.5, Qwen2.5-VL-7B-Instruct) against proprietary frontier models on a cross-cultural dataset of 2,843 videos from eight countries.
- Benchmarked local model performance on 1,359 misleading-thumbnail videos with a combined 7.6 billion views, informing failure analysis and deployment recommendations.

Safeguarding Children at Scale: Cost-Effective Multimodal LLM Detection of Inappropriate YouTube Advertising

Eman Nabeel*, Shizza Asher*, Haleema Jamil*, **Zaeem Mohtashim Khan***, Nida Tanveer, Ihsan Ayyub Qazi, Zafar Ayyub Qazi

The Web Conference (WWW) 2026 – Web4Good Track — *Equal contribution

- Co-authored a peer-reviewed paper accepted at The Web Conference (WWW) 2026, a top-tier (A*) venue.
- Re-annotated a dataset of 2,466 YouTube advertisements under updated 2025 child-safety policies ($\kappa = 0.903$).
- Proposed DAVSP, a dynamic frame-sampling pipeline reducing multimodal inference cost by $\sim 21\times$ while staying within ~ 1 F1 point of full-video analysis.
- Evaluated multimodal LLMs (video, audio, metadata, transcripts) for scalable detection of inappropriate ads on child-oriented content.

RESEARCH EXPERIENCE

LUMS

January 2025 – Present

Multi-Turn Red-Teaming of LLM Guardrails Against Malicious Prompt Escalation

- Designed and executed a multi-turn red-teaming pipeline, pairing a Gemma-4 31B adversarial "red" model against a Gemini-based target system guarded by BAGEL, a lightweight ensemble-based malicious-prompt classifier, to probe how malicious intent escalates across extended conversations.
- Converted an exploratory Kaggle notebook into a cluster-deployable evaluation script, deployed on RunPod GPU infrastructure with incremental CSV checkpointing for crash recovery and reproducibility across long-running simulations. Analyzed conversation-level attack trajectories to characterize how malicious prompts that fail single-turn guardrail checks succeed cumulatively across multi-turn exchanges.

Detecting Misleading YouTube Thumbnails Using Large Language Models

- Built a multi-modal LLM pipeline to detect misleading YouTube thumbnails across a cross-cultural dataset of 2,843 videos from eight countries, evaluating open-weight vision-language models against proprietary frontier models.
- Findings contributed to a peer-reviewed paper accepted at ICWSM 2027.

Cost-Effective Multimodal LLM Detection of Inappropriate YouTube Advertising

- Built a multimodal LLM pipeline, DAVSP, to detect inappropriate YouTube advertising on child-oriented content, reducing inference cost by $\sim 21\times$ while staying within ~ 1 F1 point of full-video analysis.
- Findings contributed to a peer-reviewed paper accepted at The Web Conference (WWW) 2026, a top-tier (A*) venue.

Hybrid Bayesian-Deep Learning Framework for Intent-Aware Prosthetic Control

- Co-authored a research proposal to develop and simulate a novel hybrid framework for robust, intent-aware prosthetic arm control, combining discriminative deep learning with probabilistic reasoning.
- Designing a modular pipeline that fuses a compact neural network for sEMG-based intent classification with a recursive Bayes filter, aiming to maintain a stable, interpretable belief distribution over user intent rather than committing to hard classifications at each timestep.
- Targeting validation on the public NinaPro dataset in a 3D simulation environment, with the hypothesis that the hybrid approach reduces spurious intent switches and improves early detection latency compared to pure neural-only baselines.

Mastani: A Multimodal Robotic Framework for Autonomous Object Retrieval

- Developed a novel robotic system on a DJI RoboMaster platform, leveraging multimodal models (Gemini 2.0) for real-time, multilingual command interpretation and autonomous object retrieval in cluttered indoor environments.
- Designed and integrated the complete software stack, enabling the robot to autonomously identify, navigate to, and retrieve specified objects based on natural language commands.

Autonomous Trash Skimming Boat

- Co-developed an autonomous boat using a YOLO-nano computer vision model on a Jetson Nano to detect, classify, and collect floating waterway debris in real time.

LUMScape

- Collaborated in a 5-person team to build "LUMScape," a 3D digital twin of the university campus in Unity, with a Node.js/Express/MongoDB backend streaming live event and facility data.

PROFESSIONAL EXPERIENCE

LUMS – Research Associate with Dr. Zafar Ayyub Qazi July 2026 – Present

- Conducting research on LLM safety and cross-cultural content moderation, extending prior work on multimodal content detection and child online safety.
- Concurrently serving as a Teaching Assistant for Distributed Systems (CS582), conducting tutorials and office hours on consensus algorithms, replication, and fault tolerance, while designing interactive browser-based educational games and contributing to course content development.

LUMS – Teaching Assistant / Head TA — Network Centric Computing, Introduction to Blockchain, Fundamentals of Computer Science. September 2024 – May 2026

- Led and coordinated TA teams for classes of 200-300+ students across Network Centric Computing, Blockchain, and Fundamentals of Computer Science, managing grading consistency, tutorial scheduling, and office hour coverage.
- Co-designed assignments, lab modules, and course content with faculty, including redesigning curriculum components for Fundamentals of CS to strengthen OOP instruction.
- Conducted weekly tutorials and office hours explaining core concepts in distributed/web programming, blockchain consensus algorithms, and smart contracts.
- Debugged and reviewed student programming assignments, provided technical feedback on code efficiency and secure transaction design, and assisted in grading exams and quizzes.

Systems Limited – VSI Data Analytics Intern June 2025 – July 2025

- Developed and automated ETL workflows using SSIS, improving data integration efficiency and pipeline reliability.
- Analyzed large datasets to identify performance bottlenecks and implemented data validation processes, gaining foundational experience in data management frameworks applicable to large-scale robotics data pipelines.

LUMS - Research Assistant with Dr Zartash Afzal May 2024 - August 2024

- Contributed to a four-person team under Dr. Zartash Afzal to redesign the 'Fundamentals of Computer Science' curriculum.
- Won a grant worth PKR 500,000 to redesign course content according to new guidelines and introduce AI tools.

HONOURS & AWARDS

Dean’s Honor List for all 4 years- LUMS	September 2022 - June 2026
Lums Learning Institute Design and Innovation Grant - 500,000PKR	June 2024
Merit Scholarship 100% – Lahore Grammar School Johar Town	September 2020 - May 2022

Skills & Interests

Programming Languages: C | C++ | Python | SQL | C# | Go | Solidity | Java | Javascript
Software & Tools: Unity | ROS | Gazebo| PyTorch | TensorFlow | OpenCV | SSIS | SSMS | Git | Kaggle | Docker
Skills: Cloud/GPU Compute | Docker | Distributed Systems | Multimodal LLM Pipelines | ETL & Data Pipelines
Interests: Gaming | Robotics | Formula 1 | Football | Reading | Writing